

Sensitivity-Driven Optimization of a Batch Cultivation DAE Model of *Mycobacterium smegmatis*

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Abstract—Tuberculosis remains a leading cause of mortality worldwide, driven by the adaptive metabolism of *Mycobacterium tuberculosis*. Studies with the surrogate *Mycobacterium smegmatis* are costly and time-consuming, motivating the use of *in silico* models. Here, a published differential-algebraic equation (DAE) model was refined by incorporating oxygen transfer and consumption, enabling simulation of oxygen-limited and hypoxic conditions relevant to granulomas. In parallel, a Python package, DAE-Model-Calibration-and-Analysis, was developed in CasADi to automate DAE formulation, simulation, sensitivity analysis, and parameter calibration using Particle Swarm Optimization. Sensitivity and Fisher Information Matrix analyses reduced the original parameter set to a subset of identifiable candidates. Calibration of $X_{\max}$, $\mu_{\max}$, and $Y_{X/N}$ improved accuracy, lowering NRMSE to 9.3% (biomass), 9.3% (glycerol), 20.2% (ammonia), and 4.9% (pH), with global NRMSE of 1.9%. Final calibrated values ($X_{\max} = 1.1537$ g/L, $\mu_{\max} = 0.133$ h⁻¹, $Y_{X/N} = 10.449$ g biomass/g N) remained physiologically consistent. The oxygen-augmented model also achieved superior information criteria (AIC = -154.4, BIC = -149.3). By reproducing hallmark responses under oxygen stress, the model offers a mechanistic tool for tuberculosis research, supporting granuloma-like simulations, experimental design, and drug or nutrient perturbation studies.

Index Terms—Differential-algebraic equations, *Mycobacterium smegmatis*, oxygen dynamics, model calibration, CasADi, sensitivity analysis, Tuberculosis, systems biology

I. INTRODUCTION

Tuberculosis (TB) remains the leading cause of death from infectious diseases worldwide [1]. In 2023, an estimated 10.8 million people developed TB, resulting in approximately 1.25

million deaths [1]. To meet the 2027 targets set by the United Nations, an annual investment of USD 22 billion is required for prevention, diagnosis, treatment, and care [1].

TB is caused by *Mycobacterium tuberculosis*, a pathogen capable of persisting in a dormant state within the host [2]. During latency, the bacterium undergoes physiological adaptations that confer phenotypic drug resistance, making it difficult to eradicate [2]. As part of the immune response, macrophages phagocytose the bacilli and initiate granuloma formation to contain the infection [2]. Within granulomas, *M. tuberculosis* experiences carbon starvation, oxygen limitation, and acidic pH, which further drive adaptive responses [2].

Because direct *in vitro* work with *M. tuberculosis* is hazardous, researchers often employ the non-pathogenic surrogate *Mycobacterium smegmatis*. This fast-growing species shares key metabolic and regulatory traits with *M. tuberculosis*, making it a cost-effective model for studying mycobacterial physiology [3]. Nevertheless, wet-lab experiments are labor-intensive and limited in scalability [4].

Computational modeling provides a flexible alternative for exploring bacterial behavior under controlled conditions, integrating diverse datasets and performing *in silico* experiments that complement laboratory research [4]. Many approaches—ordinary and partial differential equations, agent-based models and hybrid multiscale frameworks—have been used to investigate immune responses, granuloma formation, cytokine signaling and nutrient dynamics during TB infection [4].

Focusing on latency, Magombedze and Mulder [5] developed a gene-expression-based ODE model of *M. tuberculosis* under nitric oxide, hypoxia, and nutrient deprivation, reproducing transitions between replicative and non-replicative states. Ibarguen-Mondragón et al. [6] constructed an ODE model of macrophage–T cell–bacteria interactions, revealing multiple infection equilibria and bistable responses driven by immune clearance rates and bacterial competition.

Unlike host–pathogen or population-level models, this work examines the metabolic dynamics of *M. smegmatis* in batch culture. Building on a previously published DAE framework [7], the model was refined by explicitly incorporating oxygen transfer and consumption. Additionally, a CasADi-based Python package was developed to automate and simplify DAE formulation, simulation, sensitivity analysis, and parameter calibration—which are time-consuming. Local sensitivity analysis and Fisher Information Matrix evaluation guided the selection of parameters for calibration, yielding a model with improved fitting accuracy.

II. METHODOLOGY

A. Model Extension and Implementation

This work extends the dynamic model proposed in [8], which itself derives from the foundational model proposed in [7]. The present formulation preserves the following elements from [7]: it is based on Monod kinetics, considering glycerol and ammonium as limiting substrates; it introduces additional terms to improve predictive accuracy, one such term accounts for the inhibition of bacterial growth due to medium acidification, while another models the lag phase. Biomass production, substrate consumption, and metabolite formation are all described through systems of ordinary differential equations (ODEs). Furthermore, a pH estimation model is incorporated by solving an electrical charge balance, accounting for dissociated species in the medium.

In the extension developed by [8], the carbon dioxide (CO_2) production equation is modified to capture its role in pH regulation, and an additional term is introduced into the growth rate expression to represent bacterial self-inhibition at high biomass concentrations.

The model proposed in this work further expands the previous frameworks by incorporating the oxygen dynamics described by [7], including an ODE for oxygen concentration and its integration into the growth rate expression.

B. Mathematical modeling

1) *Mass balances:* The biomass concentration X (g/L) was modeled through a net growth rate that accounts for microbial proliferation and decay, where μ is the specific growth rate (1/h) and k_d is the constant specific death rate (1/h):

$$\frac{dX}{dt} = (\mu - k_d) \cdot X \quad (1)$$

The glycerol concentration C (g/L) was modeled based on its consumption for biomass formation, assuming Monod-type

kinetics. The rate of consumption is proportional to X and μ , scaled by the yield coefficient $Y_{X/C}$ (g biomass/g glycerol):

$$\frac{dC}{dt} = - \left(\frac{\mu}{Y_{X/C}} \right) \cdot X \quad (2)$$

The ammonium concentration N (g/L) was modeled analogously to the carbon source, with consumption driven by biomass growth and scaled by the yield coefficient $Y_{X/N}$ (g biomass/g ammonium):

$$\frac{dN}{dt} = - \left(\frac{\mu}{Y_{X/N}} \right) \cdot X \quad (3)$$

Carbon dioxide concentration CO_2 (g/L) was modeled as a product of biomass growth, scaled by the yield coefficient Y_{X/CO_2} (g biomass/g CO_2). Additionally, the model includes its consumption through the formation of bicarbonate, which depends on the proton concentration $[H^+]$ and contributes to pH regulation. The parameter K_{a5} (mol/L) is an equilibrium constant that represents the dissociation of carbonic acid according to acid–base reactions:

$$\frac{d\text{CO}_2}{dt} = \frac{\mu}{Y_{X/\text{CO}_2}} \cdot X - K_{a5} \cdot \frac{\text{CO}_2}{\frac{[H^+]}{K_{a5}} + 1} \quad (4)$$

The dynamics of dissolved oxygen concentration O_2 (g/L) were modeled through a differential equation that accounts for two key processes: gas-liquid transfer and microbial consumption. The transfer is described by a term driven by the difference between the saturation concentration $O_{2,\text{sat}}$ (g/L) and the actual dissolved oxygen, scaled by the volumetric oxygen transfer coefficient $k_L a$ (1/h). The consumption term reflects oxygen uptake due to biomass growth, governed by μ , X , and the biomass yield per unit of oxygen consumed Y_{X/O_2} (g biomass/g O_2). This dynamic was not included in the previously proposed models, since no oxygen limitation was observed under their experimental conditions. However, oxygen availability is known to play a critical role in systems involving biofilms and granulomas, where diffusion limitations become relevant [7]. Therefore, explicit inclusion of oxygen dynamics is essential to improve the physiological realism and predictive capacity of the model in such contexts.

$$\frac{dO_2}{dt} = k_L a \cdot (O_{2,\text{sat}} - O_2) - \frac{\mu}{Y_{X/O_2}} \cdot X \quad (5)$$

The derivation and mass balances for these equations can be found in the Supplementary Material Section A.

2) *Charge balance:* The pH of the growth medium was modeled dynamically through an electro-neutrality condition that accounts for the dissociation of relevant ionic species. To determine which components of the medium and metabolic products significantly influence pH, only species with acid dissociation constants $pK_a < 9$ were considered, as these exhibit appreciable dissociation under physiological conditions.

The following algebraic charge balance equation is solved at each time point to obtain the proton concentration $[H^+]$

(mol/L), which is the only unknown. More details in the Supplementary Material Sections B and C.

$$[H^+] = [OH^-] + [KHPO_4^-] + 3[C_6H_5O_7^{3-}] + 2[C_6H_6O_7^{2-}] + [C_6H_7O_7^-] + [HCO_3^-] - pH_{alk} \quad (6)$$

Once $[H^+]$ is determined, the pH is calculated as:

$$pH = -\log_{10}[H^+] \quad (7)$$

The specific growth rate μ is defined as a function of time and is governed by multiple biological and environmental factors. The expression integrates five multiplicative terms that account for lag phase, substrate availability, oxygen limitation, self-inhibition due to biomass accumulation, and pH-dependent inhibition.

μ_{max} is the maximum specific growth rate (1/h), t is time (h), and t_{lag} is the characteristic lag time (h), representing the initial adaptation period. The terms C , N and O are each modulated by their half-saturation constants K_C , K_N and K_O (g/L), respectively. X_{max} is the maximum biomass capacity (g/L), which models self-inhibition through logistic kinetics. The effect of extracellular pH on microbial growth is represented by the dimensionless inhibition factor $I(pH)$, which penalizes deviations from the physiologically optimal pH range.

$$\mu = \mu_{max} \left(1 - e^{-\frac{t}{t_{lag}}}\right) \cdot \frac{C}{C + K_C} \cdot \frac{N}{N + K_N} \cdot \frac{O}{O + K_O} \cdot \left(1 - \frac{X}{X_{max}}\right) \cdot I(pH) \quad (8)$$

This last term reduces the specific growth rate under both acidic and alkaline stress, and is defined by three key parameters: the lower and upper pH bounds, pH_{LL} and pH_{UL} (dimensionless), which establish the viable range for growth, and an inhibition shape constant I_{val} (dimensionless), which determines the sensitivity of the system to pH deviations. The expression used is the following:

$$I(pH) = e^{\left(I_{val} \cdot \frac{pH - pH_{UL}}{pH_{UL} - pH_{LL}}\right)^2} \quad (9)$$

3) Statistical Validation and Parameter Analysis of the Model: Model validation was conducted by comparing simulations to experimental validation data using Root Mean Square Error (RMSE), Normalized Mean Squared Relative Error (NMSRE), Mean Absolute Percentage Error (MAPE), Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) metrics. Local sensitivity analysis perturbed each parameter by $\pm 10\%$ around its nominal value and computed time-averaged normalized finite differences to identify the most influential parameters. Practical identifiability was then assessed by assembling the Fisher Information Matrix from sensitivity matrices and error covariances, inverting it to obtain parameter variances, and calculating t -values (parameter estimate divided by its standard error). Parameters with high sensitivity, low correlation and $|t| > 2$ were selected

for calibration. Normality, correlation, and comparison to experimental data plots are also generated for a visual analysis.

4) Parameter Estimation: Calibration of the refined DAE model was performed using Particle Swarm Optimization (PSO) to minimize the sum of squared errors between simulated and experimental trajectories of biomass X , glycerol C , ammonium N and pH. Guided by the prior sensitivity and FIM analyses, four parameters— X_{max} , $Y_{X/N}$, pH_{UL} and μ_{max} —were chosen as decision variables within bounds found in Supplementary Material Section H.

The PSO settings were: 50 particles, 200 iterations, cognitive coefficient $c_1 = 2.0$, social coefficient $c_2 = 1.5$, and inertia weight $\omega = 0.5$. Ten independent runs were conducted to ensure convergence stability. After each run, the optimized parameter set was applied to the full DAE system:

For each perturbed parameter set, the model was simulated and relative sensitivities were computed at every time point. The sensitivities S_{ij} for parameter j and output i were defined as the time-averaged normalized finite difference, as described in [8]:

$$S_{ij} = \frac{1}{T} \int_0^T \left| \frac{y_i^+(t) - y_i^-(t)}{2\delta \cdot y_i(t)} \right| dt \quad (10)$$

Where $y_i^+(t)$ and $y_i^-(t)$ are the model outputs obtained with the increased and decreased values of parameter j , respectively, $\delta = 0.10$ is the magnitude of the perturbation, and T is the simulation time horizon. In practice, the integral in (10) was approximated using the discrete average across all simulation time points.

For the FIM analysis, simulations are run utilizing the initial conditions given by experimental data for parameter estimation. As described in [8], the FIM is defined as:

$$FIM = \sum_{l=1}^N \mathbf{G}(t_l)^T \cdot \mathbf{Q}_l^{-1} \cdot \mathbf{G}(t_l) \quad (11)$$

Each optimized set was validated by recomputing RMSE, NMSRE, MAPE, AIC and BIC, and updating the FIM for fresh t -value estimates.

Note that pKa values, the fixed concentrations of phosphate and citric acid, and the ionic contribution term pH_{alk} were treated as constant parameters, as their values are determined by well-established chemical equilibria or experimental preparation protocols, and do not vary under the simulated conditions. Furthermore, preliminary tests confirmed that their influence on the model outputs was negligible within physiologically relevant ranges.

Initial parameter values and state concentrations used in the model correspond to those reported by [7], ensuring consistency with the original experimental setup. Initial conditions were defined separately for the fitting and validation datasets. A summary of the model parameters and constants, as well as the initial conditions for both datasets, is provided in Supplementary Material Section E.

C. Software Implementation

To simulate, calibrate and analyze the proposed dynamic model, a Python package—DAE-Model-Calibration-and-Analysis—was developed using CasADi, which provides a symbolic framework for defining and solving index-1 DAEs algebraic constraints. The workflow of the package was designed to cover the complete cycle of dynamic model analysis. A user supplies a set of parameters along with their lower and upper bounds. The package automatically generates all possible combinations of these parameters within the specified ranges. For each combination, the complete calibration workflow is executed for a user-defined number of iterations, ensuring a thorough exploration of the parameter space and increasing the robustness of the results.

The workflow for each calibration and analysis cycle goes as follows:

- 1) **System definition:** the symbolic representation of the DAE system and its algebraic constraints was automatically generated from user-supplied equations.
- 2) **Numerical simulation:** model trajectories were computed using the IDAS solver from the SUNDIALS suite. This solver provided robust handling of stiff systems and was particularly suited for resolving the algebraic equations.
- 3) **Sensitivity analysis:** parameter sensitivities were estimated by finite differences, and both numerical results and visualizations were produced. This step facilitated the identification of parameters that exerted the greatest influence on model outputs.
- 4) **Identifiability assessment:** the Fisher Information Matrix (FIM) was assembled from sensitivity data, and parameter t -values were calculated to evaluate statistical identifiability.
- 5) **Calibration:** for each parameter combination, estimation was carried out using Particle Swarm Optimization (PSO), a stochastic population-based algorithm, within the pre-defined parameter bounds. The calibration step was repeated for the specified number of iterations, allowing the assessment of convergence stability and reducing the influence of stochastic variability.
- 6) **Validation:** after each calibration, error metrics including RMSE, NRMSE, MAPE, AIC, and BIC were recomputed. Diagnostic plots were generated for each iteration, including residual analyses, correlation heatmaps, and visual comparisons between simulated trajectories and experimental validation data.

The package outputs included both numerical data in CSV files (parameter estimates, sensitivity measures, validation metrics) and graphical reports (sensitivity profiles, identifiability plots, model-versus-data comparisons). The workflow was structured to be generalizable, such that any DAE model could be analyzed by simply substituting its governing equations and parameter ranges. The complete source code and documentation were made available on GitHub under the repository

DAE-Model-Calibration-and-Analysis Package.

1) **Comparative Simulations:** Three models were used in order to compare different formulations and assess the role of oxygen availability in *Mycobacterium smegmatis* cultivation dynamics:

- 1) **Original model:** Corresponds to the formulation proposed by Dr Apiyo [7], applied with its reported optimal parameters (13 estimated parameters). This version includes additional terms in the governing equations that were not considered in the model developed in the present work.
- 2) **Modified original model:** De Witt [10] developed an extended model derived from Apiyo's, simplifying it in terms of biomass and nitrogen dynamics. It was used together with the calibrated parameters reported in that study.
- 3) **Extended model:** The formulation proposed in this work, which builds on De Witt's simplified model by explicitly incorporating oxygen dynamics. This extension follows the experimental setup described by Apiyo for oxygen dynamic, introducing oxygen transfer and consumption balances that were not originally implemented.

Specifically for the Extended model, in order to assess the influence of modifications in the DAE system and the role of oxygen availability on culture dynamics, two simulation scenarios were analyzed. The first represented an oxygen-saturated environment, while the second represented a hypoxic environment. The hypoxic case was obtained by reducing the volumetric oxygen transfer coefficient ($k_L a$), thereby mimicking poor gas-liquid transfer, and by imposing a very low initial dissolved oxygen concentration. This configuration reproduced microaerophilic conditions analogous to those encountered in granuloma environments.

The comparison between models allowed a clearer understanding of how *Mycobacterium smegmatis* behaves under different mathematical formulations and parameterizations, highlighting the effect of specific modeling assumptions. Furthermore, the comparison between oxygen-rich and oxygen-limited conditions enabled a systematic evaluation of how oxygen availability affects biomass accumulation, substrate uptake, pH dynamics, and CO_2 production, underlining the physiological importance of explicitly incorporating oxygen balances into the DAE formulation.

III. RESULTS & DISCUSSION

A. DAE system analysis python package

The implemented package successfully automated the complete calibration workflow. Parameter scanning, sensitivity calculations, and error handling were carried out without interruption, even when non-physical or non-convergent solutions were encountered. This feature ensured that complete datasets with results were produced across all simulation runs.

The resulting dataset files contain parameter estimates, their corresponding t -values, and multiple model validation metrics. Complementary graphical outputs were generated, including

sensitivity profiles, correlation plots, and comparisons between simulated trajectories and experimental data. These results confirmed that the package provided a practical and reproducible environment for DAE-based microbial growth modeling.

B. Oxygen-Limited and Hypoxic Models

The inclusion of dissolved oxygen dynamics introduced in the proposed Extended model marked changes in system behavior.

Under oxygen-limited conditions (nominal k_{La} , high initial O_2), dissolved oxygen concentrations decreased to 0.0073 g/L after approximately 40 h. This depletion was associated with slower uptake of glycerol and ammonium, delayed pH reduction, and a lower maximum specific growth rate μ compared to oxygen-replete conditions.

When oxygen was further restricted under hypoxic conditions ($k_{La} = 5 \text{ h}^{-1}$, initial $O_2 = 0.0001 \text{ g/L}$), the effects became more pronounced. The culture required approximately 55 h to reach stabilization, biomass plateaued at 1.44 g/L, and both substrate consumption and CO_2 generation slowed significantly. The maximum μ decreased to 0.060 h^{-1} , which represented a 72% reduction compared to the oxygen-replete scenario.

These outcomes are consistent with the facultative aerobic metabolism of *M. smegmatis* and replicate patterns observed in *M. tuberculosis* under hypoxic conditions in granulomas. The results highlighted the critical role of oxygen transfer in determining growth dynamics and confirmed the need to explicitly represent oxygen in predictive models.

C. Model Calibration of Extended Model

Sensitivity analysis and FIM evaluation indicated that five parameters— $X_{\max}$, $\mu_{\max}$, $Y_{X/N}$, pH_{UL} and pH_{LL} —exerted the greatest influence on model predictions as seen on Figure S14.

The parameter bounds explored were consistent with experimental data and prior initial iterations of system simulation: $X_{\max}$ in the range 1.15–1.20 g/L, $\mu_{\max}$ in the range 0.08–0.16 h^{-1} , $Y_{X/N}$ in the range 10–15 g biomass/g N and pH_{UL} in the range 10–15.

When iterating on various combinations and values of parameters, combinations without $Y_{X/N}$ failed to capture ammonia dynamics, therefore it had to be included. Similarly, when not including $X_{\max}$ biomass would grow far beyond experimental values, therefore it had to be estimated. Finally $\mu_{\max}$ allowed the slope of simulations to better represent the experimental validation data, while pH_{UL} made simulation results visually incoherent with growth dynamics, although having a better numerical validation result (results and detailed explanation on Supplementary Material section F).

Additionally, correlation analysis showed that pH_{LL} was strongly associated with $\mu_{\max}$ (Figure S15), and therefore calibration was focused primarily on $X_{\max}$, $\mu_{\max}$, $Y_{X/N}$ and pH_{UL} . Calibration within the established ranges yielded optimal fits on a second iteration, when pH_{UL} was kept fixed

on its original value and the other three parameters were estimated.

D. Model Validation and Comparison

Residual and error analyses confirmed substantial improvements. NRMSE values were reduced to 9.2854% for biomass, 9.303% for glycerol, 20.198% for ammonia, and 4.930% for pH . Compared to previous versions of the model, this configuration resulted on better validation numbers and visual representation across all states and as a whole (Tables I and II).

TABLE I
NRMSE PER VARIABLE FOR MODEL FITTING UNDER DIFFERENT PARAMETER SETS

Variable	Apiyo [7]	De Witt [10]	Base	Calibrated
Total params.	15	12	14	14
Params. calibrated	13	4	0	3
Biomass	35.686%	32.220 %	30.930 %	9.285 %
Glycerol	23.712 %	23.575 %	22.429%	9.330 %
Ammonia	27.629%	27.648 %	28.430 %	20.198 %
pH	13.399 %	15.073 %	14.230 %	4.930 %

TABLE II
MODEL VALIDATION METRICS UNDER DIFFERENT PARAMETER SETS

Variable	De Witt [10]	Base	Calibrated
AIC	-132.622	-	-154.410
BIC	-125.867	-	-149.344
RMSE	17.243 %	34.0823 %	13.464%
NRMSE	2.4778 %	4.897 %	1.934%
MAPE	3.462	10.888	3.462

In addition to improved validation metrics, all estimated parameters achieved $|t| > 2$ (Table III) confirming their statistical significance and practical identifiability. The correlation matrix of the calibrated model, considering a 95% cut-off for high correlation, indicated three independent parameters— $Y_{X/C}$, $Y_{X/N}$, and $X_{\max}$ —and three distinct groups of highly correlated parameters:

- $\{k_C, k_N, k_O, Y_{X/O_2}, pH_{LL}, I_{val}, \mu_{\max}\}$,
- $\{pH_{UL}, t_{lag}\}$,
- $\{k_d, Y_{X/CO_2}\}$.

The estimated parameters selected for calibration were distributed across different groups, indicating that they contributed independent information to the system.

TABLE III
VALUES AND T-VALUES FOR CALIBRATED PARAMETERS

Parameters	Initial value	Optimized value	t-values
$Y_{X/N}$	21.575	10.4493286300	695.610
$X_{\max}$	1.4462	1.1536938460	6805.310
$\mu_{\max}$	0.15	0.1334781790	12697.047

These findings emphasize that the explicit consideration of oxygen transfer improved predictive accuracy and mechanistic

fidelity. The sensitivity-guided calibration approach further demonstrated that robust model refinement can be achieved by focusing on a reduced subset of influential parameters.

IV. CONCLUSIONS

A differential–algebraic model of *Mycobacterium smegmatis* batch cultivation was extended to include explicit oxygen transfer and consumption dynamics. This refinement enabled accurate simulation of both oxygen-limited and hypoxic conditions, better capturing the physiological responses of the microorganism. In parallel, a Python package based on CasADi was developed to automate model construction, sensitivity analysis, parameter calibration, and validation for restricted DAE systems. The package represents a central contribution of this work, as it provides a fully integrated, reproducible, and generalizable framework for analyzing microbial systems governed by DAEs, and even other DAE systems in different areas.

Calibration guided by sensitivity and identifiability analysis confirmed that $X_{\max}$, $\mu_{\max}$, and $Y_{X/N}$ were the most critical parameters for improving model fidelity. Estimation of these parameters reduced individual NRMSE values to 9.3% (biomass), 9.3% (glycerol), 20.2% (ammonia), and 4.9% (pH), with an overall reduction of the model NRMSE to below 2%, showing improvement from past models.

The final calibrated values were $X_{\max} = 1.1537$ g/L, $\mu_{\max} = 0.133$ h⁻¹, and $Y_{X/N} = 10.449$ g biomass/g N. These values not only produced statistically significant t -statistics ($|t| > 2$) but also remained consistent with physiologically reported ranges, confirming both their robustness and biological plausibility.

The refined model successfully reproduced hallmark adaptive responses of *M. smegmatis* under oxygen stress, including delayed biomass accumulation, slower substrate uptake, and gradual pH shifts. Beyond methodological advances, the model has direct biomedical relevance: by simulating hypoxic and oxygen-limited microenvironments, it can be applied to mimic granuloma-like conditions in tuberculosis research. This provides a cost-effective *in silico* platform for exploring drug responses, latency mechanisms, and metabolic adaptations of *Mycobacterium tuberculosis* through its non-pathogenic surrogate.

Future work will focus on integrating additional stressors (e.g., nutrient deprivation, acid challenge), exploring alternative or sequential calibration strategies to reduce parameter correlations, and expanding the package with additional solvers and multiobjective optimization workflows. Together, the refined model and the developed Python package lower the barrier for advanced DAE-based modeling in microbiology, offering a tool with both engineering and biomedical applications.

DATA AVAILABILITY

The experimental datasets used for model calibration and validation were supplied by the research group in De Witt et al. [10], based on the original measurements collected by Dr. Apiyo [7].

SUPPLEMENTARY MATERIAL

Supplementary Material is available at: DAE-Model-Calibration-and-Analysis Package.

REFERENCES

- [1] WHO. "Tuberculosis." WHO. Accessed: Mar. 14, 2025. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/tuberculosis>
- [2] K. L. Devlin, D. T. Leach, K. G. Stratton, G. Lamichhane, V. S. Lin, and K. E. Beatty, "Proteomic characterization of *Mycobacterium tuberculosis* subjected to carbon starvation," *mSystems*, vol. 10, no. 3, pp. e01530-24, Apr. 2025, doi: 10.1128/msystems.01530-24.
- [3] J. A. Sundarsingh T., J. Ranjitha, A. Rajan, and V. Shankar, "Features of the biochemistry of *Mycobacterium smegmatis*, as a possible model for *Mycobacterium tuberculosis*," *J. Infect. Public Health*, vol. 13, no. 9, pp. 1255–1264, Sep. 2020, doi: 10.1016/j.jiph.2020.06.023.
- [4] D. Kirschner, E. Pienaar, S. Marino, and J. J. Linderman, "A review of computational and mathematical modeling contributions to our understanding of *Mycobacterium tuberculosis* within-host infection and treatment," *Curr. Opin. Syst. Biol.*, vol. 3, pp. 170–185, May 2017, doi: 10.1016/j.coisb.2017.05.014.
- [5] G. Magombedze and N. Mulder, "A mathematical representation of the development of *Mycobacterium tuberculosis* active, latent and dormant stages," *J. Theor. Biol.*, vol. 292, pp. 44–59, Sep. 2012, doi: 10.1016/j.jtbi.2011.09.025.
- [6] E. Ibargüen-Mondragón, L. Esteva, and E. M. Burbano-Rosero, "Mathematical model for the growth of *Mycobacterium tuberculosis* in the granuloma," *Math. Biosci. Eng.*, vol. 15, no. 2, pp. 407–428, Apr. 2018, doi: 10.3934/mbe.2018018.
- [7] Apiyo, "The effect of environmental conditions on growth and phenotype switching in *Mycobacterium smegmatis*," MEng thesis, Dept. Eng., Stellenbosch Univ., Stellenbosch, South Africa, 2020.
- [8] J. Sacher, P. Saa, M. Cárcamo, J. López, C. A. Gelmi, and R. Pérez-Correa, "Improved calibration of a solid substrate fermentation model," *Electron. J. Biotechnol.*, vol. 14, no. 5, Sep. 2011, doi: 10.2225/vol14-issue5-fulltext-7.
- [9] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi: A software framework for nonlinear optimization and optimal control," *Math. Program. Comput.*, vol. 11, no. 1, pp. 1–36, 2019, doi: 10.1007/s12532-018-0139-4.
- [10] M. de Witt, C. I. Díaz Figueroa, L. B. Lagos Silva, C. A. Sánchez Toledo, and F. Taunton Muzio, "Enhanced modeling of a *Mycobacterium smegmatis* batch cultivation," unpublished manuscript, Dept. of Chemical and Bioprocess Engineering, Pontificia Univ. Católica de Chile, 2024.
- [11] V. Patil and V. Jain, "Understanding Metabolic Remodeling in *Mycobacterium smegmatis* to Overcome Energy Exigency and Reductive Stress Under Energy-Compromised State," *Front. Microbiol.*, vol. 12, pp. 722229, Sep. 2021, doi: 10.3389/fmicb.2021.722229.
- [12] S. L. Tran, M. Rao, C. Simmers, S. Gebhard, K. Olsson, and G. M. Cook, "Mutants of *Mycobacterium smegmatis* unable to grow at acidic pH in the presence of the protonophore carbonyl cyanide *m*-chlorophenylhydrazone," *Microbiology (Reading)*, vol. 151, no. 3, pp. 665–672, Mar. 2005, doi: 10.1099/mic.0.27624-0.
- [13] I. L. Sparks, K. M. Derbyshire, W. R. Jacobs Jr., and Y. S. Morita, "Mycobacterium smegmatis: The Vanguard of Mycobacterial Research," *J. Bacteriol.*, vol. 205, no. 1, pp. e00337-22, Jan. 2023, doi: 10.1128/jb.00337-22.
- [14] J. Mäkelä, A. Papagiannakis, W. H. Lin, M. C. Lanz, S. Glenn, M. Swaffer, G. K. Marinov, J. M. Skotheim, and C. Jacobs-Wagner, "Genome concentration limits cell growth and modulates proteome composition in *Escherichia coli*," *Elife*, vol. 13, pp. RP97465, Dec. 2024, doi: 10.7554/eLife.97465.
- [15] Lide, D. and Frederikse, H. *CRC Handbook of Chemistry and Physics: A Ready-Reference Book of Chemical and Physical Data*, CRC Press, Boca Raton, FL, 1993.